NISCHAY BILIGI

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
MSAIE – BME (Research) | QPA: 4.0/4.0 — May 2027

Vellore Institute of Technology — Vellore, India
Bachelor of Technology – Biotechnology | GPA: 8.58/10.00 — July 2023

SKILLS

Programming Languages: *Advanced* - Python, R Programming; *Intermediate* - Java, C++, MATLAB
Technical Knowledge: Molecular Docking and Dynamics, Computer Aided Drug Discovery, Multi-omics, Biostatistics, Machine Learning, Natural Language Processing, Deep Learning, SQL, MLOps, Apache Spark, Kafka, Neo4j

RESEARCH EXPERIENCE

Chandra Lab, University of Pittsburgh Medical Center (UPMC) — Pittsburgh, Pennsylvania
Graduate Research Assistant — January 2026 – Present

- Leading independent computational analysis of high-throughput proteomic data from SomaScan 5000 protein panel to identify biomarkers for chronic lung disease therapeutic responses.
- Analyzing proteomic profiles from over 5000 participants in COPDGene and SPIROMICS cohorts to predict therapeutic response to lithium for emphysema and assess lung disease comorbidities.
- Directing development and application of machine learning models, differential expression analysis, and regression techniques to discover proteomic signatures associated with disease progression and treatment outcomes.

PROFESSIONAL EXPERIENCE

Healthseq Precision Medicine (HealSeq), Indian Institute of Science — Bangalore, India
Research Scientist — August 2024 – August 2025
Trainee Scientist — August 2023 – August 2024

- Collaborated with a team of 6 in projects primarily focused on biomarker discovery via multi-omics analysis, systems modelling and machine learning and formulated precise clinical scores to facilitate personalized treatments for Tuberculosis, Asthma, COPD and Breast Cancer.
- Spearheaded development of a clinical decision support system for tuberculosis prediction and monitoring, integrating radiological, genomic, clinical and socio-economic data with ML algorithms; deployed to 3 partnering hospitals and healthcare centers for clinical testing.
- Developed an all in one pulmonary disease diagnostic platform combining speech-to-text reporting, lung auscultation and cough sound analysis, and chest X-ray classification using ML, NLP, neural networks and explainable AI.
- Mentored Interns in techniques such as Machine learning and Biomarker Discovery.
- Represented Healseq as an exhibitor at Bengaluru Tech Summit, 2024 and as an Innovator at India Innovation Summit, 2025.

Chandra Lab, Department of Biochemistry, Indian Institute of Science — Bangalore, India
Intern — June 2023 – August 2023

- Refined expertise in Biochemistry, Computational Biology and Structural Bioinformatics under Dr. Nagasuma Chandra's guidance and got familiarized with algorithms such as SiteMotif, FLAPP and PocketDepth.

ACADEMIC PROJECTS

Unlocking New Avenues: Exploring Azoles Statins and Anti-cancerous Compounds Against Mucormycosis for Drug Repurposing — Vellore, India
Vellore Institute of Technology — December 2022 – April 2023

- Leveraged in-silico techniques to identify 5 drug candidates (azoles, statins, anti-cancer compounds) inhibiting Lanosterol 14-alpha demethylase and CotH3 targets of Rhizopus oryzae as economical treatments for mucormycosis.

Modification of Psilocin for Treatment of Depression in Adults — Vellore, India
Vellore Institute of Technology — September 2022 – December 2022

- Modified chemical structure of Psilocin to enhance 5HT1A/5HT2C receptor selectivity while reducing 5HT2A affinity, predicting reduced hallucinogenic effects with preserved therapeutic antidepressant function.

LEADERSHIP

Senior Core Committee Member, Biotech Research Society of India, Vellore — July 2020 – July 2022

- Directed research initiatives and organized conferences, workshops, events, and quizzes.